DH Workshop: sequence of activities

Everything that happens across the notebooks, in order, as a basis for planning. Generated from the notebooks in their current state: each notebook has a presentation part (A), a hands-on part (B) and an appendix that is not presented.

Notebook 0: Welcome	Notebook 1: IIIF in practice	Notebook 2: Europeana	Notebook 3: CLIP search
00_welcome.ipynb	*01_iiif_overview.ipynb*	*02_europeana_api_and_data.ipynb*	*03_clip_search.ipynb*
Welcome, structure of the notebooks, how to work in a notebook, setup check.	Introduction, three browser demos, IIIF in Switzerland, how it works, hands-on with image URLs.	From one platform to an aggregator: what Europeana holds, how to query it, from a record to a manifest.	Searching a collection by what images look like; the final task of building a selection in a viewer.
6 presentation slides	**15** presentation slides · **4** in the browser · **1** hands-on · appendix: **6** sections	**10** presentation slides · **2** hands-on · appendix: **7** sections	**9** presentation slides · **2** hands-on · appendix: **4** sections

Activity types

Browser	Presenter works in the browser (platforms, Mirador, developer tools).	**Explain**	Text or diagram to talk through; no code to run.
Run	Code the presenter runs and discusses; participants follow.	**Exercise**	Part B: participants edit cells and run them themselves.

Before the workshop

• Kernel **DH Workshop** selected. In each notebook the first code cell (setup) runs first; later cells depend on it. The helper code lives in *notebooks/iiif_helpers.py*, *europeana_helpers.py* and *clip_helpers.py*.
• Internet access for notebooks 1 and 2, and for Mirador, which is loaded from a CDN. Notebook 0 has a setup check cell.
• Notebook 2: optional Europeana API key in *notebooks/api-key-europeana.txt*; without it a shared demo key is used and may be rate limited.
• Notebook 3: the dataset folder (*data/DSI-Workshop*: metadata, embeddings, CLIP weights) must be present.
• Notebook 1, Demo A: replace the two placeholder manifests (EARLIER, RECENT) with the real panorama manifests.

Notebook 0: Welcome

00_welcome.ipynb · Welcome, structure of the notebooks, how to work in a notebook, setup check.

#	Slide	Type	What happens	Note	Min
1	**Welcome**	**Explain**	Three ideas that fit together (IIIF, Europeana, CLIP) and the aim: your own selection in a viewer.		
2	**How the notebooks are organised**	**Explain**	Part A presentation, Part B hands-on, Appendix: the three coloured banners.		
3	**The three notebooks**	**Explain**	Table of the three notebooks: what you see, what you do.		
4	**Working in a notebook**	**Explain**	Cells, Shift+Enter, run in order, kernel DH Workshop, restart if stuck.		
5	**Check your setup**	**Run**	One cell checks Python, packages, IIIF and Europeana servers, dataset folder.	Fix problems before notebook 1.	
6	**Ready**	**Explain**	Open notebook 1.		

Notebook 1: IIIF in practice

01_iiif_overview.ipynb · Introduction, three browser demos, IIIF in Switzerland, how it works, hands-on with image URLs.

#	Slide	Type	What happens	Note	Min
PART A · Presentation					
1	**Notebook 1: IIIF in practice**	**Explain**	Agenda of the notebook.		
2	**What is IIIF?**	**Explain**	Standard and community; open APIs. Source: Loosli (2025).		
3	**Why does it matter?**	**Explain**	Diagram: institutions, IIIF APIs, any viewer. Images stay at the institution.		
4	**Where do I find IIIF?**	**Browser**	Platforms that offer IIIF; look for the IIIF icon or manifest link.	Open the platforms in the browser.	
5	**How do I use it?**	**Explain**	Four steps: find, copy, open a viewer, paste. Compare in Mirador, workspace export.		
6	**Demo A: two sources in one viewer**	**Browser**	Two manifests from two institutions in the Mirador demo; Google Maps as third state.	Demo A manifests EARLIER / RECENT are still placeholders.	
7	**Demo B: one Book of Hours, many owners**	**Browser**	Fragmentarium reconstruction F-mp8l, one leaf, manifest in Mirador.		
8	**Demo C: a terapixel image, and its tile stream**	**Browser**	Terapixel panorama of the Battle of Murten, developer tools Network tab, Img filter.	Needs F12 on the presenter machine.	
9	**IIIF in Switzerland**	**Explain**	IIIF in Switzerland: 23 platforms, versions, consortium members. Source: Raemy (2026).	Table of platforms is in the appendix.	
10	**How it works behind the scenes**	**Explain**	Presentation API and Image API.	Live requests from here on.	
11	**A manifest**	**Run**	Fetch a Cookbook manifest, label, pages, thumbnail strip.		
12	**One image, any view**	**Run**	Anatomy of an Image API URL; four views: full, region, rotated, grey.		
13	**Deep zoom with tiles**	**Run**	Deep zoom: the same 512 px window at four scales, built from tiles; data transferred.	Ties back to Demo C.	
14	**Notes on a region of a page**	**Run**	Annotation with #xywh target, overlay; e-manuscripta transcription example.		
15	**The same manifest, in a viewer**	**Run**	The same manifest in Mirador embedded in the notebook.	Needs CDN access.	
PART B · Hands-on					
16	**Exercise: change an image by editing its URL**	**Exercise**	Change an Image API URL: rotate, mirror, grey, crop, thumbnail, combine. Stretch task and solutions.	Exercise.	

Appendix (not presented): A. The structure of a manifest in more detail · B. Following the chain from canvas to image · C. The Image API in more detail · D. Annotations: coordinates and canvas size · E. IIIF in Switzerland · F. Summary

Notebook 2: Europeana

02_europeana_api_and_data.ipynb · From one platform to an aggregator: what Europeana holds, how to query it, from a record to a manifest.

#	Slide	Type	What happens	Note	Min
PART A · Presentation					
1	**Notebook 2: Europeana**	**Explain**	From one platform to an aggregator.		
2	**How data gets to Europeana**	**Explain**	Institutions, harvesting, aggregators, IIIF (diagram).		
3	**The Europeana API**	**Run**	Search API and Record API; the index counted by media type.		
4	**Who contributes?**	**Run**	Items per institution with media-type mix.		
5	**Routes and countries**	**Run**	Providers and aggregators; countries.		
6	**Which institutions name a IIIF manifest?**	**Run**	Institutions that name IIIF manifests (lower bound).	Link to notebook 1.	
7	**A closer look at one institution**	**Run**	Profile of one institution (Wellcome Collection).		
8	**Beyond the website's search box**	**Run**	Stacked filters: keyword, images, reusable, years, IIIF.		
9	**What one record contains**	**Run**	Fields of one record; print a sample record.		
10	**From a record to a IIIF manifest**	**Run**	From a record to a manifest URL; image service check on 20 records.	Closes the loop to notebook 1.	
PART B · Hands-on					
11	**Exercise: find an object with a IIIF manifest**	**Exercise**	Search with keyword, period, IIIF switch and theme; note the number of a result.	Exercise.	
12	**Exercise: from a result to its manifest**	**Exercise**	Number to manifest URL, image service check, copy URL, embedded Mirador.	Exercise. Stretch: add it in the Mirador demo.	

Appendix (not presented): A. How the data gets to Europeana in more detail · B. The Search API in more detail · C. Counting institutions in more detail · D. Another institution · E. Beyond the search box in more detail · F. The IIIF filter and the manifest check · G. Summary

Notebook 3: CLIP search

03_clip_search.ipynb · Searching a collection by what images look like; the final task of building a selection in a viewer.

#	Slide	Type	What happens	Note	Min
PART A · Presentation					
1	**Notebook 3: Searching by what images look like**	**Explain**	Searching by content with embeddings.		
2	**How the dataset was made**	**Explain**	How the shared dataset was made: vectors and links, no images.	Instructor-side pre-processing.	
3	**The collection**	**Run**	Load the collection; random sample; top institutions and countries.	Needs the data folder.	
4	**CLIP: text and images in one space**	**Run**	CLIP figure (credited): text and image encoder; load the model.	Needs the model weights.	
5	**Search with text**	**Run**	Text query to vector, similarity, top 12.		
6	**Restricting the search with a mask**	**Run**	Mask: only images before 1900 (country / institution alternatives).	The mask demo.	
7	**Where do the results come from?**	**Run**	Institutions in the top 100 vs the collection.		
8	**Search with an image**	**Run**	Image as query: more like this.		
9	**Look closely: a larger version, on request**	**Run**	Larger version or detail via IIIF service, or original download; data compared.	Ties back to notebook 1.	
PART B · Hands-on					
10	**Exercise: build your own selection**	**Exercise**	Choose a theme, write a query, optional mask, note numbers of 3 to 5 results.	Exercise.	
11	**Exercise: from your picks to a viewer**	**Exercise**	Numbers to manifest URLs, check, embedded Mirador with all picks, text block for the shared pad.	Final task. Instructor collects URLs, show_mirador_many.	

Appendix (not presented): A. The dataset folder · B. Writing queries · C. Search with your own image · D. Summary

The sequence in one line

0 · Welcome	Three ideas → how the notebooks are organised → working in a notebook → setup check
1 · IIIF	What / why / where / how to use → demos A, B, C in the browser → Switzerland → manifest, image URL, tiles, annotations, Mirador → hands-on: edit an image URL
2 · Europeana	Data flow → the API → institutions, routes, countries → IIIF-linked → one institution → stacked filters → one record → record to manifest → hands-on: search, pick, manifest
3 · CLIP	Dataset → collection → CLIP → text search → mask → bias → image search → look closely → hands-on: theme, picks, shared viewer, paste URLs into the pad

Points to think about

• The notebooks contain no timings. The **Min** column is empty for your own estimates.
• Hands-on parts: notebook 1 has one exercise (edit an image URL), notebook 2 has two (search and filters, then from a result to its manifest), notebook 3 has two (theme query, then picks to a shared viewer). Everything else is presenter-led.
• Notebook 1 has the longest presentation part (introduction, three browser demos, Switzerland, technical walk-through). The three Browser slides in the introduction and demos are the parts that happen outside the notebook.
• The final task of notebook 3 produces a list of manifest URLs for the shared pad. The instructor can open all of them in one viewer with *show_mirador_many([...])*. Mirador can also export and import a workspace, which is not yet part of the notebook.
• Each notebook has an appendix with the detail that is not presented (tables, code walk-throughs, references).